AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_B14FC11ECE
Taxa	Mamenchisaurus
Clade	Sauropod
PBDB Records	24
Mass (regression)	10213.9 kg (Campione & Evans 2012)
Mass (volumetric)	9643.5 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	11.480 m³
Validity Score	1.0000 / 1.0000
TCI	0.7093 (partial)
Axiom I Entropy	0.9488
Axiom II λ■	0.1991 (symmetry 100%)
Axiom III Uniformity	0.7007
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	70.4% Measured / 29.6% Inferred
Elements	27
CoG (m)	[1.002, 2.032, 0.0]
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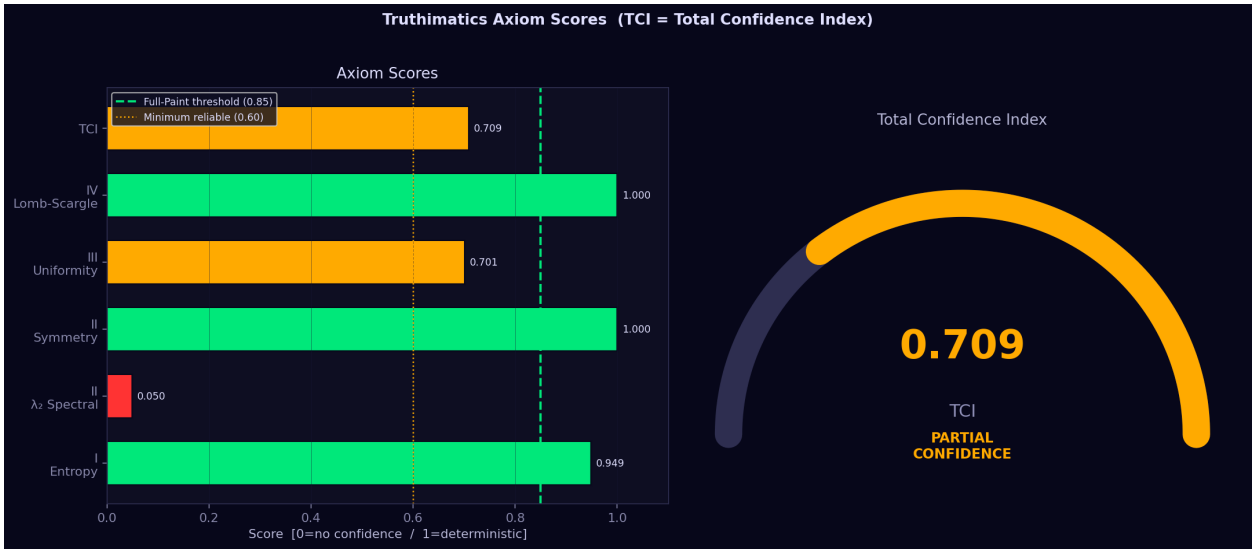
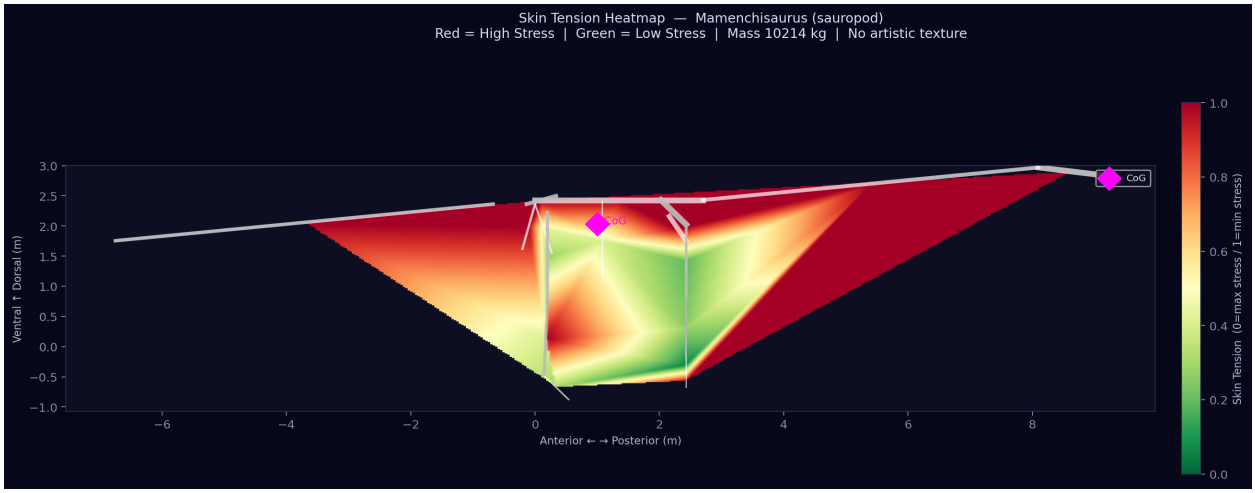
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

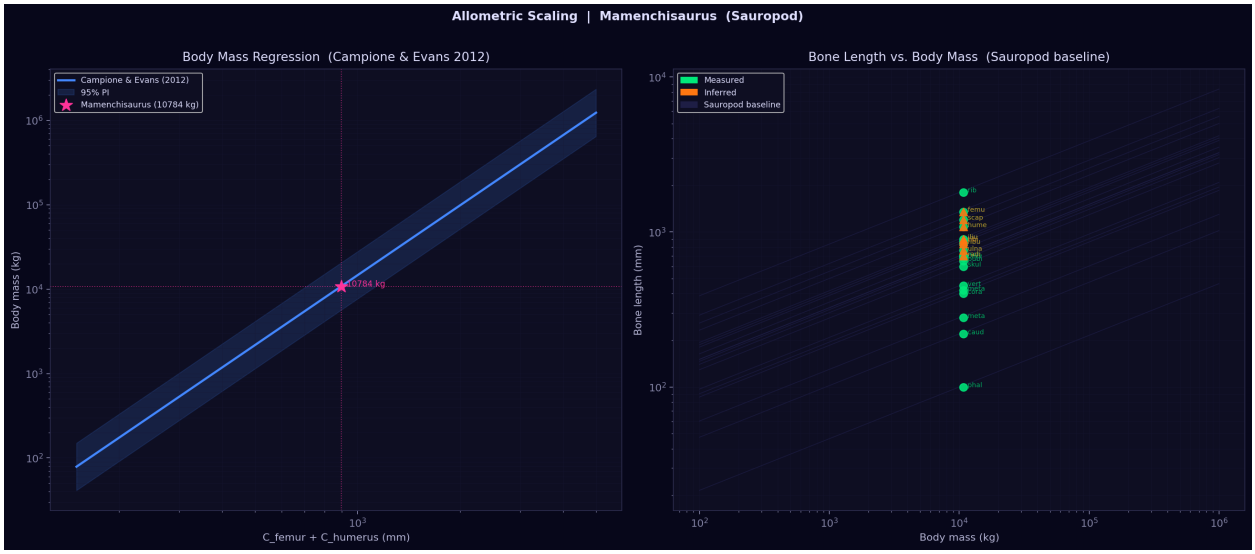
Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
caudal_vertebra	0.677	1.131	1.808	15.000	OK
cervical_vertebra	0.548	2.549	3.097	15.000	OK
coracoid	0.351	0.768	1.118	15.000	OK
femur	2.607	31.057	33.664	5.050	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	3.707	42.112	45.819	3.710	WARNING
ilium	1.566	7.710	9.276	15.000	OK
ischium	1.523	9.717	11.239	15.000	OK
metacarpal	14.682	69.887	84.569	2.010	WARNING
metatarsal	9.788	22.488	32.275	5.267	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	2.707	21.387	24.094	7.056	OK
radius	3.034	34.176	37.211	4.569	OK
rib	0.418	24.672	25.090	6.776	OK
scapula	0.179	0.839	1.018	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	4.350	20.703	25.053	6.786	OK
ulna	2.707	31.259	33.965	5.005	OK
vertebra	0.696	1.428	2.124	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.64 m	Femur + tibia + metatarsal
Estimated Speed	12.8 km/h (3.55 m/s)	Gait: Walk | Fr=0.488
Stride Length	4.884 m	Alexander (1976) / Thulborn (1990)
Bite Force	2502.0 N (255.0 kgf)	Sauropod regression (Whitlock 2011)
Skull Length	600.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3607	Circumference / length (dimensionless)
Limb Strength Index	3.2e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	18.79 m	Allometric from femur length
Est. Neck Length	7.89 m	Allometric from femur length
Est. Tail Length	7.89 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
caudal_vertebra	220.0	180.0	2855.14	3564.1	Measured	0.88	15.00
cervical_vertebra	680.0	200.0	10895.04	13600.3	Measured	0.88	15.00
coracoid	400.0	250.0	10013.83	12500.3	Measured	0.88	15.00
femur	1350.0	155.0	12991.44	16217.2	Measured	0.88	5.05
femur_right	1350.0	155.0	12991.44	16217.2	Symmetry	0.84	5.05
fibula	840.0	55.0	718.45	753.5	Measured	0.88	15.00
fibula_right	840.0	55.0	718.45	753.5	Symmetry	0.84	15.00
humerus	1080.0	130.0	7310.89	9126.2	Measured	0.88	3.71
humerus_right	1080.0	130.0	7310.89	9126.2	Symmetry	0.84	3.71
ilium	900.0	200.0	14419.91	18000.4	Measured	0.88	15.00
ilium_right	900.0	200.0	14419.91	18000.4	Symmetry	0.84	15.00
ischium	700.0	120.0	4037.57	5040.1	Measured	0.88	15.00
metacarpal	280.0	65.0	176.53	145.2	Measured	0.88	2.01
metatarsal	420.0	80.0	1076.69	1344.0	Measured	0.88	5.27
phalanx	100.0	55.0	36.49	28.3	Measured	0.88	15.00
pubis	650.0	90.0	2108.91	2632.6	Measured	0.88	7.06
radius	700.0	85.0	2025.80	2528.8	Measured	0.88	4.57
radius_right	700.0	85.0	2025.80	2528.8	Symmetry	0.84	4.57
rib	1800.0	50.0	1802.49	2250.0	Measured	0.88	6.78
scapula	1200.0	350.0	58881.30	73501.5	Measured	0.88	15.00
scapula_right	1200.0	350.0	58881.30	73501.5	Symmetry	0.84	15.00
skull	600.0	350.0	29440.65	36750.8	Measured	0.88	15.00
tibia	870.0	120.0	5018.13	6264.1	Measured	0.88	6.79
tibia_right	870.0	120.0	5018.13	6264.1	Symmetry	0.84	6.79
ulna	760.0	90.0	2465.80	3078.1	Measured	0.88	5.00
ulna_right	760.0	90.0	2465.80	3078.1	Symmetry	0.84	5.00
vertebra	450.0	300.0	16222.40	20250.4	Measured	0.88	15.00